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ORIGINAL PAPER

In-situ classification of football sports surfaces: leveraging machine learning for enhanced surface analysis

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Abstract
Sports turf surfaces, including natural turfgrass and synthetic turf, are complex systems with many parameters influencing their performance. This study aims to classify sports turf surfaces using data collected from a bespoke testing device, fLEX, which calculates seven separate metrics related to sports surface performance in both an acceleration format (designed to simulate an athlete accelerating) and deceleration format (designed to simulate an athlete decelerating). Sixty-eight collegiate and professional sports surfaces across the USA and UK were tested, covering a range of climates and field constructions. Surfaces were classified as cool-season, warm-season, or synthetic turf. After data preprocessing, including outlier removal and imputation, two machine learning models, decision tree and random forest, were trained and tested on the dataset. Feature importance was assessed using mutual information, revealing that recoil distance and maximum vertical force were the most critical variables for classification. The decision tree model achieved an accuracy of 84% for acceleration and 79% for deceleration, while the random forest model performed slightly better, with accuracies of 89% and 83%, respectively. Both models demonstrated low overfitting risk, with a minimal difference between training and testing accuracies. Misclassifications were analysed, highlighting the complexity of surface characteristics and potential for improving classification accuracy. The high performing models suggest that the fLEX testing device is an appropriate tool to classify the surfaces, and that unique characteristics exist within each surface category. Collectively, these findings represent a step toward advancing our understanding of the complexity of sports turf surfaces.

Keywords Sport surface classification · Synthetic turf · Natural turfgrass · Machine learning · Traction testing · Player-surface interaction · Artificial turf · Soccer

1 Introduction

Sports turf surfaces are multifaceted systems broadly divided into natural turfgrass and synthetic turf. These categories each have further nuanced differentiations. Natural turfgrass systems, for example, can vary in turfgrass species and rootzone composition [1]. Species choice is typically reliant on the climatic condition. Cool-season turfgrass surfaces found in moderate climates are usually comprised of *Poa pratensis* and *Lolium perenne* while hot climates typically utilise warm-season species *Cynodon dactylon* (X *C. transvaalensis*), *Paspalum vaginatum*, or *Zoysia* spp [1]. The rootzone for turfgrass systems are typically either sand-based at the collegiate and professional levels of sport and native at parks, recreational areas, and high school levels of sport [1]. Whereas synthetic turf systems have diverse fibre types, infills, and shock pads [2], which are traditionally determined by the manufacturer in response to sport-specific performance requirements, regulatory standards, and cost considerations [2, 3]. Evidence suggests that these parameters can alter the performance of a sport surface system [4–7]. For example, McNitt et al. [8] found higher values of traction on *Poa pratensis* compared to *Lolium perenne* surfaces. On synthetic turf, the sand-to-crumb-rubber ratio impacts playing-surface hardness, defined in this study as the peak deceleration during an impact with the surface, not material hardness in the material engineering sense (i.e., resistance to local plastic deformation) [9]. Further

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complexities are added due to the field managers' maintenance decisions for the system. Mowing height, thatch level, and soil moisture impact traction characteristics of a natural turfgrass system [8, 10, 11]. Synthetic turf maintenance, such as infill replacement and brushing, can also impact field performance [12]. Therefore, understanding the effects of sports surface composition on player-surface interactions is essential for developing a comprehensive assessment of sports surface performance, enabling field managers to make data-driven decisions regarding surface maintenance and player interactions.

Player-surface interactions can be quantified by using athletes or indirectly via mechanical tests that simulate athlete movement. Player-surface mechanical tests offer the advantage of simulating key aspects of human athlete biomechanics without the need for athlete involvement, ensuring a more reproducible and reliable assessment of surface performance while mitigating variations between individual athletes [13]. These tests are commonly classified into two categories: vertical impact and traction. Impact tests function by dropping a weighted missile onto the surface and measuring the forces (traditionally with an accelerometer) imparted onto the missile after it impacts the surface. While there are many variations of this test, the two most common are the Clegg Impact Hammer which measures surface hardness and the Advanced Artificial Athlete which measures force reduction, energy restitution, and vertical deformation [14, 15]. Surface hardness is the maximal deceleration (i.e., maximum negative acceleration) from impacting the surface while force reduction is a surface hardness normalized to concrete [15]. Energy restitution is a measurement of how much energy is returned to the testing device from the surface and is commonly calculated from the area under the unloading phase of the force–deformation curve [15]. Finally, vertical deformation is the measure of how much the surface deforms from the impact of the missile and is most frequently calculated by double-integrating the acceleration over the duration of the impact. Higher levels of surface hardness positively correlate with an increased risk of lower extremity injury [16]. Likewise high levels of energy restitution and vertical deformation may also negatively impact the safety and performance of an athlete on the surface [17, 18].

Traction tests quantify the amount of grip from an athlete on a given surface, capturing a complex interaction between the sports surface, the football boot's outsole, and its studs. Many factors beyond the surface design and condition impact traction, such as the type of studs used, stud placement, forces applied during testing, and the testing device operator [10, 11, 19]. Rotational traction tests quantify the torque returned to the athlete when a studded test foot is rotated on the surface. Typically associated with athletic cutting movements, rotational traction has a predictive capability for the safety of a given surface [20]. Rotational traction measurements can be influenced by inconsistencies in how the operator rotates the device, which may increase measurement error [19]. In contrast, translational traction is the acceleration and deceleration ability of an athlete on a given surface. Translational traction is measured by striking a cleated foot that is fitted with force sensors linearly across a surface and measuring the peak force the foot receives [21, 22]. Traditionally, translational traction is measured with large pneumatic or motor driven devices which are either lab bound or difficult to haul [21, 22]. In both types of traction testing, there is a point where insufficient traction causes slippage, reducing athletic performance. Conversely, excessive traction increases the risk of lower extremity injuries [16].

Football is played in diverse climatic environments worldwide and relies on the Fédération Internationale de Football Association (FIFA's) quality assurance standards to maintain surface uniformity [15]. Despite FIFA's efforts, the multitude of surfaces encountered in the sport, ranging from natural turfgrass to synthetic turf and hybrid systems, presents challenges in standardization of performance. Traditional univariate analyses of surface testing fail to capture the complex, multidimensional interactions between players and surfaces [23–26]. To address this gap, the present study employs a bespoke testing device that measures multiple mechanical properties of the playing surface, including hardness, energy return, and traction. Prior studies indicate that these metrics differ meaningfully among cool-season, warm-season, and synthetic surfaces [6, 13, 27–29], providing a mechanistic basis for distinguishing surface types. By integrating these measurements through multivariate machine learning techniques, we aim not only to classify surfaces with high accuracy but also to identify the key mechanical properties that differentiate them. This approach offers a data-driven understanding of how surfaces vary and how their unique characteristics may influence performance, thereby informing both future research and practical surface management.

2 Methods

Sixty-eight collegiate and professional stadia and training grounds (fields) were tested across the United States of America and United Kingdom covering a wide range of climates, field constructions, and management practices. These surfaces were classified during testing as either synthetic turf (26 of 27 were crumb rubber) or natural turfgrass (2 of 20 cool-season surfaces were a stitched hybrid system). Hybrid turfgrass systems were classified as natural turfgrass



Fig. 1 The fLEX device used to collect data

systems, as the majority of coverage was still natural turfgrass, and research has suggested they still perform similar to natural turfgrass [27–30]. If the surface was natural turfgrass it was further classified as a warm- or cool-season surface based on the predominant turfgrass species at the time of testing. Multiple data points were collected across the surface during testing, with a median of 24.5 (minimum = 3; maximum = 99; median absolute deviation = 14.5) samples per field for acceleration format and 17.5 (minimum = 2; maximum = 90; median absolute deviation = 10) for deceleration format; however, both the number and location of samples varied according to the field manager's preference, with the most common approach being a grid and others using transects across the surface, all collected within a short time frame (~ 60 min) on a given field. The acceleration format was designed to simulate an athlete's linear acceleration, where the device records traction generated as the posterior forefoot studs drive backward into the surface to provide propulsive grip. Conversely, the deceleration format simulates an athlete's linear deceleration, with the device capturing the braking resistance created as the anterior studs at the toe and outer forefoot bite into the surface to resist forward momentum [31].

2.1 fLEX testing device

The testing methodology employed a custom-designed device, fLEX, specifically engineered to replicate athlete biomechanics during acceleration and deceleration on various playing surfaces (Figs. 1 and 2) [32]. A 27 cm (US men's size 10), 3D-printed foot was propelled toward the surface with a heel-elevated at a 15-degree angle, simulating running conditions validated by prior methodologies [21, 33] impacting the surface at 1.60 ± 0.37 m s^{-1}. The propulsion mechanism consisted of a calibrated spring system capable of simulating a range of athlete body masses (23–159 kg) to mimic real-world impacts. The foot, affixed to a spring-driven carriage, impacted the surface at a 22° angle to replicate the interaction between an athlete's foot and the playing field. For this study, the fLEX device was configured to represent the mean mass of senior professional association football players, 77.2 kg [34].

The fLEX device integrated sensor technology, including a tri-axial force sensor (261B03, PCB Piezotronics, Depew, NY) mounted at the rigid ankle joint, with a manufacturer-rated overall measurement uncertainty of ± 1%, linearity values of 0.05%, 0.08%, and 0.4%, and sensitivities of 0.28, 0.28, and 0.05 mV N^{-1} for the X-, Y-, and Z-axes, respectively. Additionally, a uniaxial position sensor (MPA-323THTP0, Sick Sensor, Minneapolis, MN) was affixed to the foot carriage to measure foot–surface interaction along a 22° angle, with a manufacturer-rated resolution of ≥ 0.06 mm, linearity of 0.5 mm, and repeatability of ≥ 0.1 mm. A Savitzky–Golay filter was applied to the position and force signals to reduce high-frequency noise while preserving the true temporal and amplitude characteristics of the measured responses [35]. This filtering approach was selected because it performs local polynomial regression within a moving window, allowing effective smoothing without distorting peak magnitudes, rise times, or phase relationships that are critical for accurately characterizing force–displacement behaviour. All filters used first-order polynomials, with a window length of 51 for the position sensor and 31 for the triaxial force sensor. The selected polynomial order and window length were chosen to be consistent with the

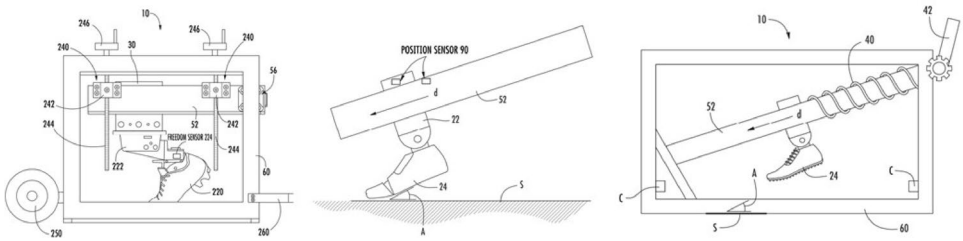


Fig. 2 Schematics of the fLEX device. The left panel illustrates the full device and the placement of the three-degrees-of-freedom force sensor. The centre panel shows the position sensor and foot impact angle. The right panel depicts the spring-propelled foot carriage and its approach angle to the surface (Reproduced from patent number US 12,379,306 B2)

expected signal bandwidth and sampling frequency, ensuring attenuation of noise components above the mechanically relevant frequency range while maintaining fidelity of transient features. Validation testing confirmed that the chosen parameters minimized signal variance without altering key metrics such as peak forces in the vertical and horizontal orientation, displacement, or timing parameters, supporting their suitability for repeatable application across trials.

From these sensor data, seven key mechanical variables were calculated. Cleat penetration depth is the distance the foot travels into the surface after initial surface contact, measured from the point at which the force sensor detects the impact with the surface, indicated by increasing force values, to the farthest point of foot extension into the surface along a 22° angle (Figure S1). Maximum displacement is the total foot travel from the initial firing position to its farthest point of foot extension, measured along a 22° angle using the position sensor (Figure S1). Recoil distance is the distance the foot travels back toward the firing mechanism after surface impact, measured from the point of maximum displacement into the surface to the maximum recoil position as the carriage moves back toward the spring and firing position, using position sensor data (Figure S1). Medial force is the peak side-to-side (mediolateral) force at the ankle joint while no rotation is applied to the foot; during early testing medial force was found to vary based on field type. Horizontal force is the peak forward–backward force at the ankle. Shear force is calculated as the resultant magnitude of medial and horizontal forces, and peak vertical force represents the maximum up–down force exerted on the cleat. These parameters were selected as proxies for critical surface performance metrics, including surface hardness (peak vertical force), energy restitution (recoil distance), and traction (cleat penetration depth, maximum displacement, and shear force).

2.2 Data preprocessing

Prior to any dataset preprocessing, the sample size for the acceleration test was 2,279 and 1,445 for the deceleration test, with each data point representing a single measurement at a given field location. The difference in data points reflects that the testing was conducted opportunistically rather than specifically for this study, and field access constraints limited the number of deceleration measurements. All data preprocessing, model fitting, and model evaluation were conducted in Python 3.11 [36]. Data preprocessing consisted of removing a total of 137 (6% of total samples) samples for acceleration and 131 (9% of total samples) for deceleration using an Isolation Forest with 1,000 estimators via the 'IsolationForest' function in scikit-learn v1.4 [37, 38], to further clean the dataset from extraneous values which were attributed to sensor errors. Isolation Forest is an anomaly detection algorithm that identifies outliers by recursively partitioning the data, isolating observations that deviate significantly from the overall distribution. Outliers were removed to address sensor errors that occurred during field testing, as their inclusion could compromise the robustness and accuracy of the analysis.

Following outlier removal, missing values caused by testing device malfunction were imputed using a K nearest neighbour imputation method (KNNImputer; scikit-learn v1.4) with three neighbours corresponding to the centres of the three possible surface classifications (warm-, cool-season, and synthetic turf) groups, ensuring completeness in the dataset [37, 39]. Subsequently, the data were partitioned into 80% training data and 20% testing data using the 'train_test_split' (scikit-learn v1.4), with a stratification based on the classification group [37]. The portioning of the dataset al.lows for the development and assessment of models based on independent datasets.

2.3 Feature importance

A mutual information feature selection (mutual_info_classif; scikit-learn v1.4) was fit to the training dataset to assess parameter importance in surface classification [37]. Mutual information is a non-parametric feature selection method that quantifies how much information each predictor variable provides about the target variable. In this context, it measures the reduction in uncertainty about the sports surface type achieved by knowing the value of a given predictor [40, 41]. A higher mutual information value indicates a stronger association between the predictor and the outcome variable, suggesting the parameter offers greater insight into surface type classification. This step was performed prior to model building to identify and prioritize the most relevant variables for classification.

2.4 Model fitting

Two separate tree-based models were fit to the data, a decision tree and random forests. These methods were chosen due to their non-parametric nature while maintaining interpretability. The decision tree, a single predictive model that recursively splits the data into branches based on feature values, was fitted to the training dataset with a 'balanced' class weighting to account for class imbalance [42]. An exhaustive grid search (GridSearchCV; scikit-learn v1.4) was used to tune the decision tree's hyperparameters [37]. The hyperparameters tuned were maximum tree depth from 1 to 6 and minimum number of samples per leaf from 10 to 50 to help prevent overfitting of the models. The highest scoring hyperparameters were then used to fit the decision

tree and the decision tree was tested using the testing data. A random forest model was also fitted, leveraging an ensemble of decision trees bootstrapped from the training dataset, in contrast to the decision tree model, which relies on a single predictive tree. The random forest's hyperparameters were set to a maximum depth of 6 and a minimum of 10 samples per leaf. The number of estimators hyperparameter was initially set to 1000 but was adjusted after an out-of-bag error analysis. The out-of-bag analysis consisted of plotting the out-of-bag error over the number of estimators and the value where the error stabilizes was chosen.

2.5 Model evaluation and performance

All models' performances were evaluated using precision, recall, specificity, f1-score, and overall accuracy. Precision is the accuracy of a positive prediction, in the given context it is the ability for the model to correctly identify (true positive) the proportion of surface's types that were correctly predicted from all the positive predictions (true positive + false positive). Recall, or sensitivity, is the model's ability to correctly classify the surface (true positive) against all the surfaces in the specific type (true Positive + false Negative). The F1-score is a balance of precision and recall as expressed in Eq. (1).

$$\text{F1-Score} = 2 * \frac{(Precision * Recall)}{(Precision + Recall)} \quad (1)$$

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The overall accuracy is the model's ability to correctly classify the playing surface (true positive + true negative) over all instances. Furthermore, the test model evaluations were then compared across the model to evaluate the ability performance of the various models. A confusion matrix was used to visualize misclassified parameters. Misclassifications were then more thoroughly explored by field to determine why they were misclassified.

3 Results

3.1 Acceleration model mutual importance

Mutual importance analysis revealed the contribution of various parameters to identifying the type of surface tested (Fig. 3). Recoil distance was the most important variable, with a mutual importance score of 0.38 followed by maximum vertical force, which had a score of 0.34 (see Supplementary Table S3 for median and median absolute deviations and Supplementary figure S4 for histogram of the seven fLEX parameters by surface type).

3.2 Acceleration model decision tree results

GridsearchCV determined the best hyperparameters of the model with a maximum depth of five and minimum samples per leaf of 10 (Figure S2). The decision tree analysis exhibited an overall test dataset accuracy of 84% which differed by 2% points compared to the training dataset's accuracy (Table 1). The precision of the three surface categories (cool-season, synthetic turf, and warm-season) were 87, 89, and 80%, respectively. The cool-season, synthetic turf, and warm-season categories had a 75, 76, and 95% recallability. The F1-score, or harmonic mean of precision and recall, was 80% for cool-season, 82% for synthetic turf, and 86% for warm-season. A total of 20 fields (29.4%) had one or more data point misclassified including four (5.9%) of them

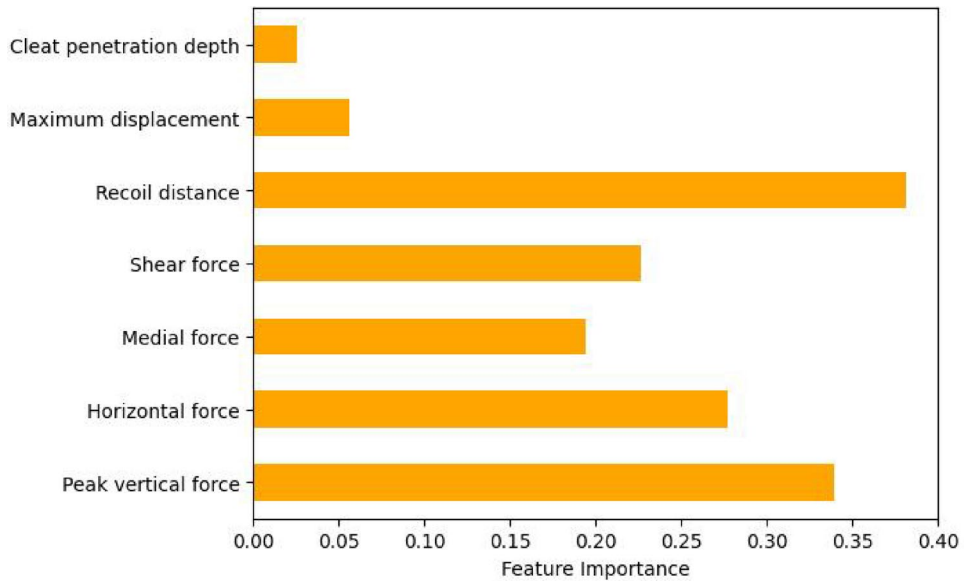


Fig. 3 Feature importance rankings for acceleration using mutual importance scores. The higher the feature importance value, the more important the variable is in sport surface classification

Table 1 Classification report from the acceleration format for the decision tree and random forest models

	Decision tree				Random forest			
	Precision	Recall	F1-Score	Support	Precision	Recall	F1-score	Support
Testing								
Cool-season	0.87	0.75	0.80	114	0.90	0.82	0.86	114
Synthetic turf	0.89	0.76	0.82	129	0.94	0.81	0.87	129
Warm-season	0.80	0.95	0.86	186	0.85	0.98	0.91	186
Accuracy			0.84	429			0.89	429
Macro Avg	0.85	0.82	0.83	429	0.90	0.87	0.88	429
Weighted Avg	0.84	0.84	0.83	429	0.89	0.89	0.88	429
Training								
Cool-season	0.94	0.79	0.82	456	0.87	0.83	0.85	456
Synthetic turf	0.87	0.86	0.87	513	0.92	0.86	0.89	513
Warm-season	0.87	0.90	0.88	744	0.88	0.94	0.91	744
Accuracy			0.86	1713			0.89	1713
Macro Avg	0.86	0.85	0.86	1713	0.89	0.88	0.88	1713
Weighted Avg	0.86	0.86	0.86	1713	0.89	0.89	0.89	1713

Table 2 Misclassification rates by field for both acceleration and deceleration formats with both decision tree and random forest models

		Decceleration										Acceleration									
		Decision Tree					Random Forest					Decision Tree					Random Forest				
Field Surface	Field Type	% Misclassified	Cool	ST	Warm	Total	% Misclassified	Cool	ST	Warm	Total	% Misclassified	Cool	ST	Warm	Total	% Misclassified	Cool	ST	Warm	Total
Field 1	Cool-season	94.4	1	0	17	18	88.9	2	10	6	18	100.0	0	0	10	10	100.0	0	5	5	10
Field 2	Synthetic turf											72.7	0	3	8	11					
Field 3	Synthetic turf	100.0	0	0	10	10	100.0	0	0	10	10	30.0	6	14	0	20					
Field 4	Synthetic turf											80.0	7	2	1	10	80.0	7	2	1	10
Field 5	Synthetic turf											49.1	17	27	9	53	79.2	33	11	9	53
Field 6	Warm-season																27.3	3	0	8	11
Field 7	Synthetic turf	100.0	4	0	0	4	100.0	4	0	0	4										
Field 8	Warm-season	40.0	1	1	3	5	100.0	2	3	0	5										
Field 9	Warm-season	70.0	3	4	3	10	90.0	4	5	1	10										
Field 10	Warm-season						100.0	5	1	0	6										
Field 11	Cool-season	28.6	5	0	2	7						27.6	55	7	14	76					
Field 12	Warm-season	55.0	11	0	9	20	55.0	11	0	9	20	100.0	20	0	0	20	45.0	8	1	11	20
Field 13	Warm-season	75.0	15	0	5	20	45.0	9	0	11	20	100.0	20	0	0	20					
Field 14	Warm-season																100.0	11	0	0	11
Field 15	Cool-season	95.5	1	0	21	22															
Field 16	Synthetic turf											32.4	0	23	11	34					
Field 17	Synthetic turf	100.0	0	0	2	2	50.0	1	1	0	2										
Field 18	Synthetic turf											90.9	0	1	10	11	100.0	0	0	11	11
Field 19	Warm-season	50.0	2	2	4	8	50.0	1	3	4	8										
Field 20	Synthetic turf											70.0	0	3	7	10	100.0	0	0	10	10
Field 21	Synthetic turf	33.3	0	2	1	3	33.3	2	0	4	6										
Field 22	Synthetic turf											28.0	0	18	7	25	32.0	0	17	8	25
Field 23	Synthetic turf	100.0	3	0	0	3	100.0	3	0	0	3										
Field 24	Cool-season											100.0	0	14	42	56	96.4	2	2	52	56
Field 25	Warm-season	30.0	0	3	7	10	60.0	1	5	4	10	50.0	0	5	5	10	100.0	0	10	0	10
Field 26	Warm-season						57.1	1	3	3	7										
Field 27	Synthetic turf	29.9	21	54	2	77															
Field 28	Synthetic turf	100.0	3	0	0	3	100.0	1	0	2	3										
Field 29	Synthetic turf	90.0	9	1	0	10	90.0	9	1	0	10	54.5	5	5	1	11	27.3	2	8	1	11
Field 30	Warm-season											36.1	10	3	23	36					
Field 31	Cool-season						70.4	8	0	19	27										
Field 32	Warm-season											40.0	0	4	6	10	40.0	0	4	6	10
Field 33	Warm-season	33.8	26	1	53	80															
Field 34	Cool-season											100.0	0	3	11	14	100.0	0	1	13	14
Field 35	Warm-season	45.2	15	4	23	42															
Field 36	Synthetic turf											72.7	0	3	8	11					
Field 37	Cool-season	60.6	13	20	0	33	78.8	7	24	2	33										
Field 38	Synthetic turf	80.0	2	1	2	5	80.0	4	1	0	5	71.4	5	2	0	7	71.4	5	2	0	7

being cool-season, 11 synthetic turf (16.1%), and five (7.4%) warm-season surfaces (Table 2). Cool-season surfaces were misclassified as a warm-season surface 16 times (3.7%) and as synthetic turf seven times (1.6%; Fig. 4). Synthetic turf surfaces had 18 cases (4.2%) of warm-season misclassification and eight cases (1.9%) of cool-season misclassification. Warm-season surfaces were misclassified as synthetic turf 13 (3%) times and cool-season seven times (1.6%).

3.3 Acceleration model random forest results

The overall accuracy of the testing dataset for the acceleration format was 89% with no difference compared to the training dataset (Table 1). The cool-season category had a precision of 90%, recall of 82% and F1-score of 86%. The synthetic turf category had a precision, recall, and F1-score of 94, 81, and 87%. The warm-season category's precision,

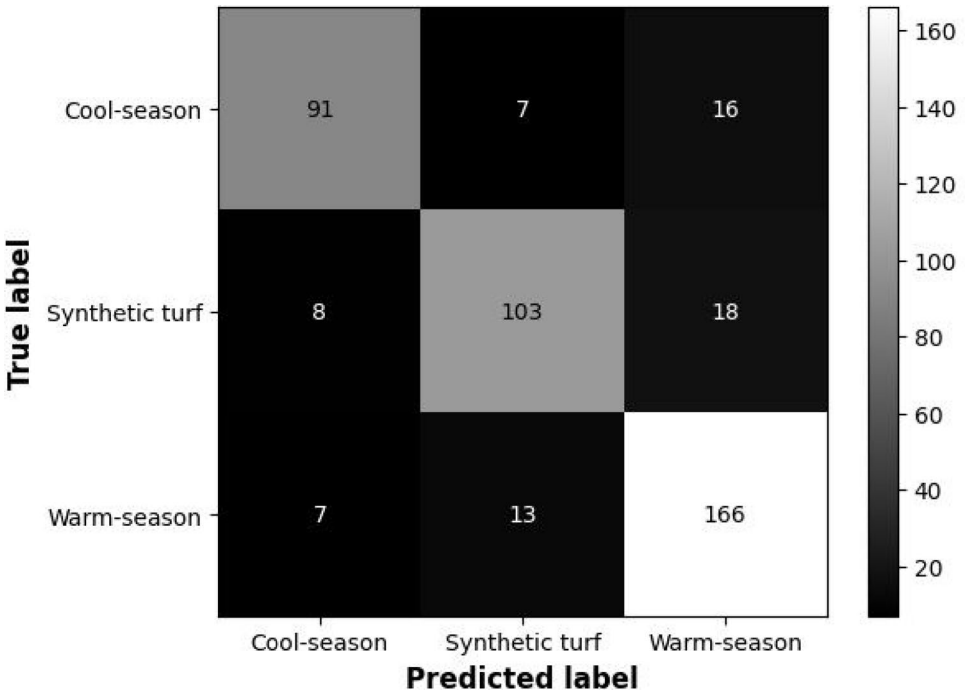


Fig. 4 Confusion matrix for the decision tree model classifying acceleration format into cool-season turfgrass, synthetic turf, and warm-season turfgrass surfaces. The matrix displays true labels on the y-axis and predicted labels on the x-axis, with values indicating the number of instances for each classification. Higher values along the diagonal represent correctly classified instances, while off-diagonal values indicate misclassifications. The grayscale shading corresponds to the number of instances, with lighter shades representing higher counts

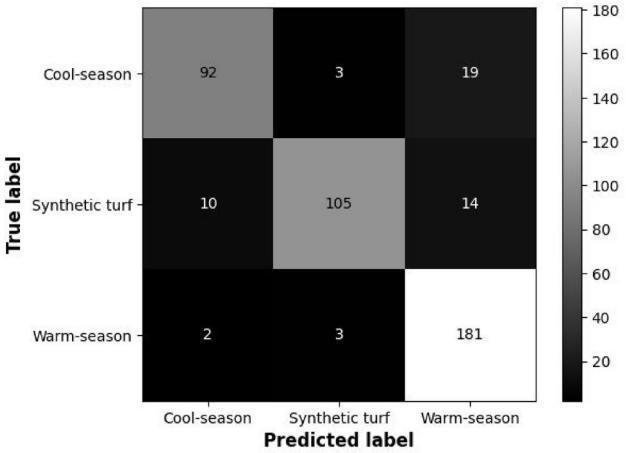


Fig. 5 Confusion matrix for the random forest model classifying acceleration format into cool-season turfgrass, synthetic turf, and warm-season turfgrass surfaces. The matrix displays true labels on the y-axis and predicted labels on the x-axis, with values indicating the number of instances for each classification. Higher values along the diagonal represent correctly classified instances, while off-diagonal values indicate misclassifications. The grayscale shading corresponds to the number of instances, with lighter shades representing higher counts

recall, and F1-score were 85, 98, and 91%. Fifteen fields (22.1%) had at least one instance of the surface being misclassified with three cool-season (4.4%), seven synthetic turf (10.3%), and four warm-season fields (5.9%; Table 2). Cool-season surfaces were misclassified as synthetic turf three times (0.7%) and warm-season surfaces 19 times (4.4%; Fig. 5). Whereas for synthetic turf surfaces, the category type had 24 cases (5.6%) of misclassification with 10 cases (2.3%) of misclassifying as a cool-season surface and 14 (3.3%) as a warm-season surface. The warm-season surfaces had two cases (0.5%) of misclassification of cool-season surfaces and three cases (0.7%) of misclassification as a synthetic turf surface.

3.4 Deceleration model mutual importance

The mutual importance analysis revealed and ranked the parameters based on their significance in determining the type of playing surface (Fig. 6). Rebound and shear force were the most important variables with a score of 0.36. Horizontal force had an importance score of 0.33, while medial force scored 0.24 (see Supplementary Table S3 for median and median absolute deviations and Supplementary figure S5 for histogram of the seven fLEX parameters by surface type).

3.5 Deceleration model decision tree results

GridsearchCV determined the best hyperparameters of the model were a maximum depth of five and minimum samples per leaf of 12; the best decision tree is exhibited in Figure S3. The overall test accuracy for the decision tree was 79%, 6% lower than the training dataset's accuracy (Table 3). The precision for cool-season, synthetic turf, and warm season categories were 80, 93, and 84%. Whereas recall for the three category types were 86, 86, and 83%. The harmonic average (f1-score) was 83, 90, and 84% for cool-season, synthetic turf, and warm-season categories, respectively. Eighteen fields (26.4%) had one or more misclassifications

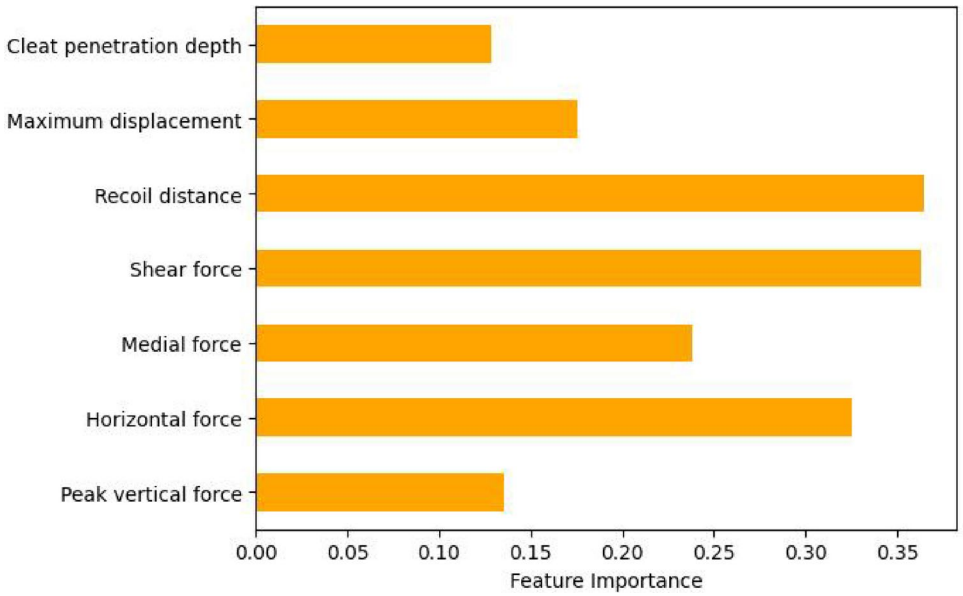


Fig. 6 Feature importance rankings for deceleration using mutual importance scores. The higher the feature importance value the more important the variable is in sport surface classification

Table 3 Classification report from the deceleration format for the decision tree and random forest models

	Decision tree				Random forest			
	Precision	Recall	F1-Score	Support	Precision	Recall	F1-Score	Support
Testing								
Cool-season	0.70	0.86	0.77	91	0.76	0.85	0.80	91
Synthetic turf	0.86	0.81	0.78	80	0.87	0.78	0.81	80
Warm-season	0.87	0.80	0.84	92	0.87	0.87	0.87	92
Accuracy			0.79	263			0.83	263
Macro Avg	0.81	0.79	0.80	263	0.83	0.83	0.83	263
Weighted Avg	0.81	0.79	0.80	263	0.83	0.83	0.83	263
Training								
Cool-season	0.80	0.86	0.83	365	0.84	0.89	0.86	365
Synthetic turf	0.93	0.86	0.90	318	0.90	0.90	0.90	318
Warm-season	0.84	0.83	0.84	368	0.90	0.85	0.87	368
Accuracy			0.85	1051			0.88	1051
Macro Avg	0.86	0.85	0.85	1051	0.88	0.88	0.88	1051
Weighted Avg	0.85	0.85	0.85	1051	0.88	0.88	0.88	1051

of data points within the surface with four (5.8%) of the fields being a cool-season turfgrass, nine (13.2%) being a synthetic turf, and nine warm-season fields misclassified (13.2%; Table 2). The misclassification of cool-season surfaces occurred with synthetic turf (five cases; 1.9%) and warm-season surface (eight cases; 3.0%) (Fig. 7). Synthetic turf surfaces had 23 (8.7%) total misclassifications with 20 (7.6%) misclassifying the surface as a cool-season surface and three (1.1%) warm-season surfaces. The warm-season surfaces were misclassified as cool-season surfaces in 14 instances (5.3%) and synthetic turf in four instances (1.5%).

3.6 Deceleration model random forest results

The overall accuracy of the random forest model was 83% which was 5% points lower than the training dataset (Table 3). The cool-season category had a precision of 76%, recall of 85% and F1-score of 80%. Synthetic turf category had a precision, recall, and F1-score of 87, 78, and 81%, respectively. For the warm-season category, the precision, recall, and F1-score were 87, 87, and 87%, respectively. Nineteen fields (27.9%) had at least one data point misclassified with three cases (4.4%) for a cool-season field, eight (11.8%) synthetic turf fields, and eight (11.8%) warm-season fields (Table 2). Similar to the decision tree, the cool-season surfaces were most frequently misclassified as a warm-season surface (nine cases; 3.4%) and then synthetic turf surface (five cases; 1.9%) (Fig. 8). Synthetic turf had 19 instances (7.2%) of misclassification with the surface being misclassified as cool-season surface 16 times (6.1%) and three instances (1.1%) of misclassification as a warm-season surface. The warm-season surface had eight cases (3.0%) of cool-season misclassification and four cases (1.5%) of synthetic turf misclassification.

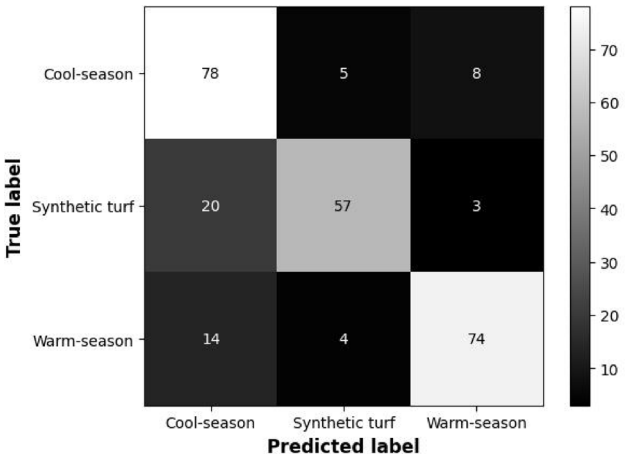


Fig. 7 Confusion matrix for the decision tree model classifying deceleration format into cool-season turfgrass, synthetic turf, and warm-season turfgrass surfaces. The matrix displays true labels on the y-axis and predicted labels on the x-axis, with values indicating the number of instances for each classification. Higher values along the diagonal represent correctly classified instances, while off-diagonal values indicate misclassifications. The grayscale shading corresponds to the number of instances, with lighter shades representing higher counts

4 Discussion

Sports turf surfaces are complex entities with many parameters which can impact how the surface performs [13]. Our goal was to classify surfaces only by broad type (cool-season, warm-season, or synthetic), despite the substantial within-type variation in species composition and construction features that can affect performance metrics [4–9, 12, 30, 43]. The accuracy of both the decision tree and random forest models in the acceleration and deceleration formats suggests that surface type was classified relatively successfully, with all models exceeding 75% accuracy.

From the parameter importance analysis, recoil distance—defined as the amount of energy returned to the testing device—was the most influential parameter. This high importance likely reflects its ability to distinguish between natural turf systems (cool- and warm-season) and synthetic turf, which is engineered to provide energy return to athletes [4, 43], similar to a trampoline effect. While acceleration and deceleration formats had many similarities in feature importance, the main difference was acceleration favoured the maximum vertical force parameter, more akin to a surface hardness or force reduction measurement, over the shear force (which is more similar to traditional traction measurement) parameters. Evidence from the decision tree supports this finding as recoil distance is a major contributor to splitting natural turfgrasses and synthetic turf (Figure S2). The second most important parameter varied by testing format: acceleration favoured maximum vertical force, a hardness-related metric, while deceleration prioritized shear force, which represents the combined medial and translational forces at the ankle. Notably, maximum vertical force in acceleration served as the initial node, distinguishing cool-season turf from warm-season surfaces.

Building on this, the acceleration and deceleration format further underscored the role of recoil distance as a key differentiator between natural and synthetic turfgrass. From a mechanistic perspective, the importance of recoil

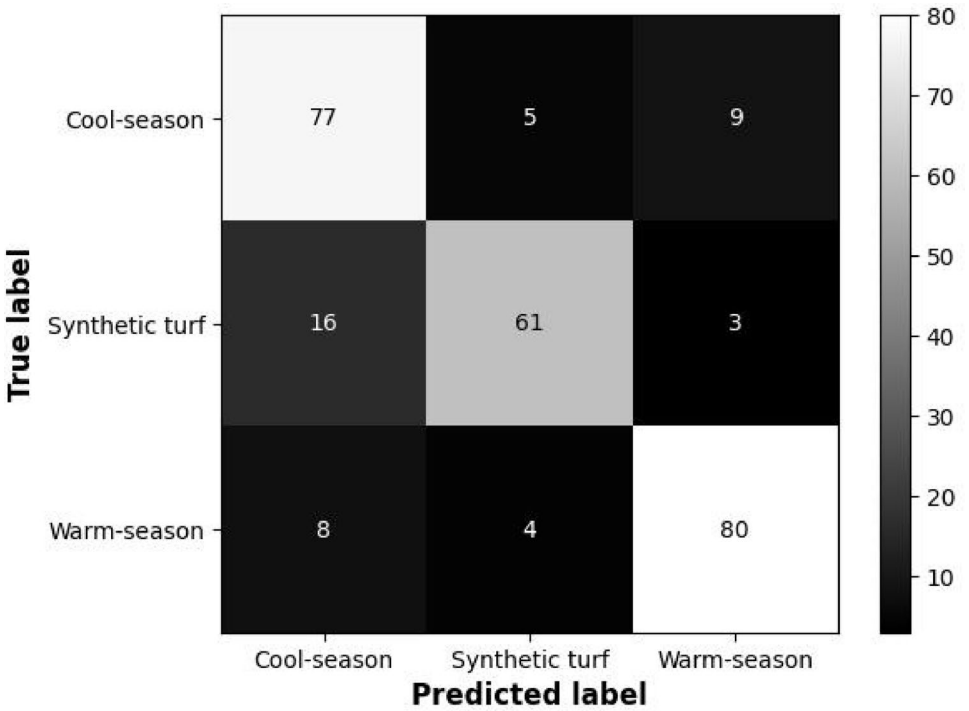


Fig. 8 Confusion matrix for the random forest model classifying deceleration format into cool-season turfgrass, synthetic turf, and warm-season turfgrass surfaces. The matrix displays true labels on the y-axis and predicted labels on the x-axis, with values indicating the number of instances for each classification. Higher values along the diagonal represent correctly classified instances, while off-diagonal values indicate misclassifications. The grayscale shading corresponds to the number of instances, with lighter shades representing higher counts

is logical given its ability to distinguish point-elastic from area-elastic systems. Synthetic turf is typically designed to act as a point-elastic surface, returning localized energy to the athlete, whereas natural turfgrass behaves more like an area-elastic system, distributing force more broadly [4, 44]. Shear force, which captures both side-to-side (mediolateral) and forward-backward (anteroposterior) contributions, offers a more complete picture of dynamic traction, whereas horizontal force aligns more closely with traditional static testing protocols [22, 45–47]. The prominence of these variables may stem from structural differences between turfgrass types—namely, the presence or absence of stolons [8]. Furthermore, traction metrics also helped differentiate synthetic turf, likely due to differences in the mechanical properties among various manufacturers [48, 49]. The distinction between warm- and cool-season turfgrasses can also be explained by the greater maximum vertical force observed in warm-season grasses, which is driven by the presence of both stolons and rhizomes [30]. In contrast, cool-season grasses, which often lack these structures, may yield a less firm playing surface.

Overall, all models demonstrated accuracy greater than 75%, indicating they performed better than random assignment (~ 33%). The models showed minimal risk of overfitting, with training and test accuracy differences within 6% [50]. Random forest models outperformed decision trees by approximately 4–5% on the test dataset. However, this improvement must be considered alongside the reduced interpretability of ensemble methods compared to single decision trees. When comparing acceleration and deceleration data collection formats, both decision tree and random forest models for acceleration showed slightly better accuracy (5% and 6%, respectively). It is important to note that the acceleration format had a larger sample size (166 test samples, 662 training samples) compared to the deceleration format.

Thirty-eight fields either had some or all of the data points misclassified. Acceleration format had 22 total (32.3%) unique fields with misclassification (20 (29.4%) decision tree and 15 (22.1%) random forest models), with synthetic turf surfaces being the most frequently misclassified. Whereas deceleration format had a total of 24 (35.3%) unique fields misclassified. Declaration format had 10 (14.7%) warm-season and 9 (13.2%) synthetic turf fields misclassified. Regardless of model or testing format, synthetic turf had the highest number of misclassifications (92 surfaces; 22.0%) with cool-season turfgrass having 72 (17.6%) misclassifications and warm season with 55 (9.9%). The higher rates of misclassification for both surfaces within the field in synthetic turf is hypothesized to be due to variance in sports performance caused by different combinations of synthetic turf components from various synthetic turf manufacturers [48, 49, 51].

Ultimately, the scope of this manuscript is to assess the model's ability to classify sports turf surfaces, which has limited viability for practitioners. While field mangers could employ similar models to help aid decision-making for altering the surface performance to better align with desired field characteristics (e.g., converting the performance of a synthetic turf surface to more of a cool-season turfgrass) [52], the study's greater contribution lies in clarifying which surface parameters define each surface category. For example, recoil distance effectively differentiates synthetic turf from natural turfgrass and could be managed through turf density or infill depth to optimize energy return and reduce injury risk [48]. Additionally, the differences observed between acceleration and deceleration phases highlight how distinct athletic manoeuvres require specific surface properties, as seen through the differences in parameter importance. These findings suggest that tailoring surface characteristics to the demands of various sports and movements could enhance both performance and player safety, providing valuable insights for scientists and field managers.

The presented models are not generalizable outside of the study population (i.e., professional and collegiate level sports fields) due to the nature of machine learning methods, as these models are bult using data-driven methods and not from physics-driven methods. The study did not control for temperature, humidity, soil water content, or infill depth (for synthetic turf) and future work should incorporate tighter control of these factors, as they may influence fLEX measurements. Each data point in this study represents a single strike on a field, treated as independent observations rather than repeated measures. While ensemble methods such as random forests are relatively robust to correlated data, repeated strikes within a field may introduce field-specific effects. To mitigate this, hyperparameters were carefully chosen to reduce overfitting, but we acknowledge that this remains a limitation of the study. However, these results provide a useful foundation for understanding the factors that drive differences between natural turfgrass and synthetic turf. Therefore, future studies could build upon these findings by moving from predictive approaches to theory-driven, causal methods to achieve a more complete understanding of the underlying processes responsible for differences between surfaces. The models classified surfaces at a broad level—cool-season, warm-season, or synthetic turf—but sports turf surfaces vary widely beyond these three categories thus further studies should consider additional factors such as level of play (e.g., parks and recreation, high school, college), cleat type, rootzone composition (e.g., sandy loam, loam, clay loam), turfgrass species, and turfgrass density. Additionally, investigating species variation within natural

turfgrass categories, as well as fibre type or manufacturer differences in synthetic turf, could provide deeper insights into how these major categories can be further classified based on more specific definitions of the sport surface.

5 Conclusion

This study demonstrated that sports turf surfaces can be effectively classified into cool-season turfgrass, warm-season turfgrass, and synthetic turf from the fLEX testing device, achieving 79–89% classification accuracy, exceeding the 33% accuracy expected under random classification. The high classification accuracy of decision tree and random forest models underscores the relevance of the measured parameters in differentiating surface types. Beyond classification, these findings provide insight into key surface characteristics that influence performance, offering potential applications for optimizing maintenance practices and surface design. Recoil provided the highest level of information for distinguishing between natural turfgrass and synthetic turf, while peak vertical force was the best predictor for differentiating between cool- and warm-season turfgrasses. This research enhances the understanding of sport surface mechanics and establishes a foundation for future studies aimed at refining mechanical testing methods to better reflect player-surface interactions.

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Author contributions C.B. conceptualized the study. J.S. provided funds to conduct the research. C.B. and K.D. collected the data. C.B. performed the initial data analysis, with E.F. contributing to further analysis. All authors contributed to data interpretation. J.S. and K.D., with partial input from C.B., developed the testing device used in the study. C.B. drafted the initial manuscript, and K.W. and J.S. provided initial critical revisions. All authors reviewed and approved the final manuscript.

Data availability Due to the nature of in-situ data collection within privately operated stadiums, the datasets used in this study are not publicly available. Data may be shared by the corresponding author upon reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

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